

Python

Released in the 1990s, Python is one of the most popular computer programming languages in the world. It takes a bit longer to learn than Scratch, but can be used to build just about anything.

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◀ 120–121 Types of programming language

Scratch

136–137 ▶

Why Python?

Created by Dutch programmer Guido van Rossum, Python is a text-based programming language. It is extremely versatile and can be used to make many different types of program, such as apps, games, and websites. Python is a great language for getting started with computer programming, and is used by many schools and universities to teach coding. Here are its most important features:

Python is **named after** the popular **British comedy series *Monty Python's Flying Circus***.



Simple and easy to learn

A simple and minimalistic language, Python is extremely beginner-friendly. The code is written in a combination of words, numbers, and punctuation. Its easy syntax allows beginners to focus on learning programming concepts without having to worry about too many details.



Free and open source

Python is an example of a FLOSS (free/libre and open source software), which means that it can be freely distributed, its source code can be read and changed, and its code can be used in new programs. The Python community even encourages people to contribute code, documentation, and resources.



Portable

Python is extremely flexible and can run on a wide variety of hardware platforms and operating systems. Programming languages with these qualities are called “portable”. From Windows to Mac, Linux, PlayStation, and more, Python works everywhere. Its interface looks the same and the programs behave the same way on each platform.



Embeddable

Embeddable with C or C++ encoding, Python allows its users to improve their code with scripting functions. The code can be inserted into an application to provide a programmable interface. It can also be used as a scripting language for building large applications.



Extensive library

Python's greatest strength is its standard library, which supports many common programming tasks, such as connecting to web servers, reading and modifying files, and searching text with regular expressions. It also contains built-in modules that make it easier and quicker to build programs.



Great support

Python provides comprehensive and well-written documentation to its users. It has a guide to getting started, a reference section to explain what things mean, and a lot of example code. Its active support community makes sure Python projects have detailed and easy-to-understand technical documentation.